

Noken²

Implicit implicit contextual token learning

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Abstract

Composing implicit token models is a way to update queries in a query-key-value attention mechanism. Coherent refinement is achieved when Renyi α -entropy is reduced. This approach jointly learns query, key, value embeddings and query updates, coordinated by concatenated implicit models.

Noken²

Input target token t_i From context $[t_1, \dots, t_m] \in \mathbb{R}^{r \times m}$

In implicit model (1) learn $V, Q, {}^s K, {}^t K$ and s_i

Given s_i from (1), in implicit model (2) learn

$V', Q', {}^s K', {}^t K$ and r_i . Given $q_i = Q t_i$:

Let $\alpha_{ij} = \text{softmax}_j \text{ score}(q_i, {}^s k_j)$

and $j_i^\alpha = \arg \max_j \alpha_{ij}$, where score is a similarity function.

Let $h_i = \sum_{j=1}^m \alpha_{ij} v_j \in \mathbb{R}^n$

Let $\beta_{ij} = \text{softmax}_j \text{ score}(h_i, {}^t k_j)$

and $j_i^\beta = \arg \max_j \beta_{ij}$.

Similarly in (2) given $q'_i = Q' s_i$

define α'_{ij} , h'_i , and β'_{ij} .

Output token $O_{i+1} = \begin{cases} t_{j_i^{\beta'}} & \text{if } E(\{\beta'_{ij}\}) < E(\{\beta_{ij}\}) \\ t_{j_i^\beta} & \text{otherwise} \end{cases}$

So $[t_1, \dots, t_m]$ yields $[o_2, \dots, o_{m+1}]$

for token prediction or seq-to-seq learning.

Summary:

In (1) q_i has a similarity distribution $\{\beta_{ij}\}$

over keys $\{t_{kj}\}$, likewise in (2) q'_i has a

similarity distribution $\{\beta'_{ij}\}$ over $\{t'_{kj}\}$.

If $E_{q'_i} < E_{q_i}$, that is the average Renyi α -entropy

is reduced, we accept the more certain update q'_i

to replace q_i and then continue refining queries

until entropy is no longer reduced, or is sufficiently

small or learning implicit models becomes expensive.

$$[t_1, \dots, t_m] \ni t_i \rightsquigarrow q_i \rightsquigarrow s_i \rightsquigarrow q'_i \rightsquigarrow t_{j\beta'_i}$$

□

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